

Fig. 1: Semiconductor with extra electrons

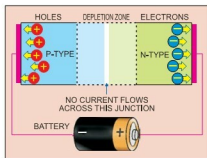


Fig. 2: Semiconductor with extra holes

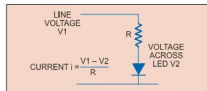


Fig. 3: Circuit of an LED

efficient with higher luminous flux (measured in lumen) per unit electrical input power (measured in watt).

The most recent record is just over 300-lm/W, which can be compared to 16-lm/W for regular light bulbs and close to 70-lm/W for fluorescent lamps. As about one fourth of the world's electricity consumption is used for lighting purposes, LEDs will contribute to saving Earth's resources. Materials consumption is also diminished as LEDs last up to 100,000 hours, as compared to 1000 hours for incandescent bulbs and 10,000 hours for fluorescent lights.

Electronics of LED

In the case of LEDs, the conductor material is typically aluminium-gallium-arsenide (AlGaAs). In pure AlGaAs, all atoms bond perfectly to their neighbours, leaving no free electrons (negatively-charged particles) to conduct electric current. In doped material, additional atoms change the balance by either adding free electrons or creating holes where electrons can go. Either of these alterations makes the material more conductive.

A semiconductor with extra electrons is called n-type material, since it has extra negatively-charged parti-

cles. In n-type material, free electrons move from a negatively-charged area to a positively-charged area.

A semiconductor with extra holes is called p-type material, since it effectively has extra positively-charged particles. Electrons can jump from hole to hole, moving from a negatively-charged area to a positively-charged area. As a result, holes themselves appear to move from a positively-charged area to a negatively-charged area.

A diode consists of a section of n-type material bonded to a section of p-type material, with electrodes on each end. This arrangement conducts electricity in only one direction. When no voltage is applied to the diode, electrons from the n-type material fill holes from the p-type material along the junction between the layers, forming a depletion zone.

In a depletion zone, the semiconductor material is returned to its original insulating state—all holes are filled, so there are no free electrons or empty spaces for electrons, and charge cannot flow.

To get rid of the depletion zone, you have to get electrons moving from the n-type area to the p-type area and holes moving in the reverse direction. To do this, you connect the n-type side of the diode to the negative end of a circuit and the p-type side to the positive end. The free electrons in the n-type material are repelled by the negative electrode and drawn to the positive electrode. The holes in the p-type material move the other way. When the voltage difference between elec-

trodes is high enough, electrons in the depletion zone are boosted out of their holes and begin moving freely again. The depletion zone disappears and charge moves across the diode.

Circuit design

In an electronics circuit, an LED behaves very much like any other diode. LEDs are often used to indicate the presence of a voltage at a particular point and are often used as a supply-rail indicator.

When used in this fashion, there must be a current-limiting resistor placed in the circuit. The resistor's value should be calculated to give the required level of current. For many devices, a current of around 20mA is suitable, although it is often possible to run them at a lower current. If less current is drawn, the device will obviously be dimmer.

When calculating the amount of current drawn, voltage across the LED itself may need to be taken into consideration. The voltage across an LED in its forward-biased condition is just over a volt, although the exact voltage is dependent upon the diode and, in particular, its colour. Typically, a red one has a forward-voltage of just below two volts, and it is around 2.5 volts for green or yellow.

Great care must be taken not to allow a reverse bias to be applied to the diode. Usually, LEDs have a reverse breakdown of very few volts. If breakdown occurs, then the LED is destroyed. To prevent this from happening, an ordinary silicon diode can be placed across the LED in reverse direction to prevent any reverse bias being applied.

Development of LEDs

LED, in electronics, is a semiconductor device that emits infra-red or visible light when charged with an electric current. LEDs operate by electroluminescence, a phenomenon in which emission of photons is caused by electronic excitation of a material. The material used most